JMO – 2000

Max Mark – 100 Time – 3 Hrs

Instructions:

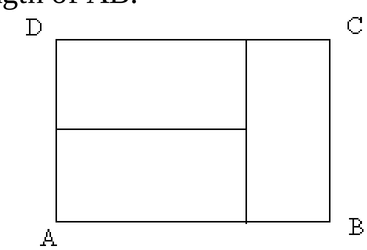
- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer all questions.

1. .

(a) Find out the digit in 100's place of the sum: $1+11+111+\ldots\ldots\ldots+\underbrace{11\ldots11}_{30\,digits}$

(b) In a contest to guess number of cricket balls kept in a basket, Anita guessed 25, Suprava guessed 31, Sureya guessed 29, Rama guessed 27 and Chandana guessed 23. Two guesses were wrong by 2 and two guesses were wrong by 4, the other guess was correct. Determine the number of balls in the basket.

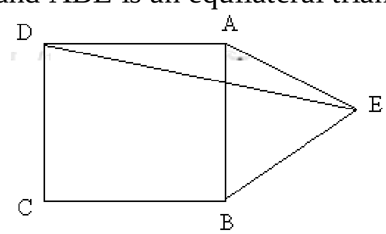
(c) The three smaller rectangles in the figure are congruent (i.e. of the same shape and size). The length BC = 1 cm. Find out the length of AB.



(d) When I opened my mathematics book there are two pages which faces me and the product of two page number is 1806. What are the two page numbers?

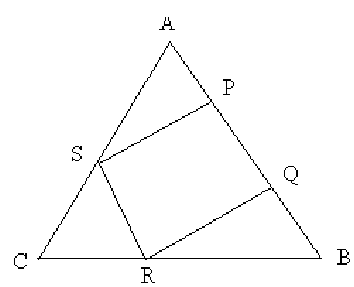
(e) Each day Bimala ate 20% of the mangoes that were with him at the beginning of the day. At the end of the day he had 32 mangoes. How many mangoes were there in the beginning of the first day?

(f) In the figure, ABCD is a square and ABE is an equilateral triangle. Find $m\angle AED = ?$

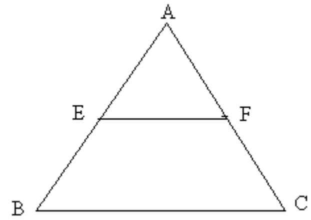


(g) For which integral value of m, $\dfrac{1}{2m^2+15}$ will be closest to 0.01?

(h) The quadrilateral PQRS is inscribed in the triangle ABC. Which has a larger perimeter: triangle ABC or quadrilateral PQRS?



(i) In the triangle ABC, AB = AC and E, F are point on AB and AC respectively. EF || BC, show that EC = BF.



(j) Find out the remainder and quotient when 1111111111 is divided by 1111.

2.

(a) An article was sold at a profit of 12%. If the article would have been purchased 10% less and sold Rs. 6 more, then the profit would have been 25%. What is the purchase cost of the article?

(b) A and B starts from two places X and Y respectively, which are 60 km apart. If A and B travel at uniform speed of 80 km/h and 20 km/h and return back immediately to their respective places after reaching Y and X respectively, then how often and at what distance from X, shall they meet?

3.

(a) Show that $2^{48}-1$ is divisible by 9.

(b) A shopkeeper cheats 10% both the seller while buying and the purchaser while selling by using a defective balance. What is his % of profit?

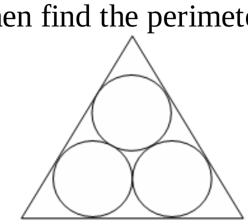
4.

(a) Put brackets to make the statement true: $5-2\times1+4\div6=5$.

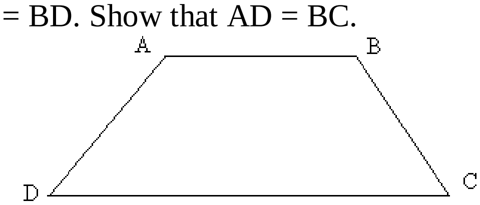
(b) If * represents an operation $a*b=b^a+a^b$, then find (2*3)*2.

5.

(a) In the figure, the three equally large circles touch each other and they touch the edge of a triangle. If the radius of each circle is 3 cm, then find the perimeter of the triangle.



(b) In the trapezium ABCD, AC = BD. Show that AD = BC.



6.

(a) In a classroom desks are arranged in a straight row. John is in third row from the front and fourth from back. He is also third from the left and fifth from the right. How many desks are there in the classroom?

(b) Between Kalia and Kaluram one lies on Monday, Tuesday and Wednesday and tell truth on all other days. The other lies on Thursday, Friday and Saturday and tells the truth on all other days. At noon, the two had the following conversation:

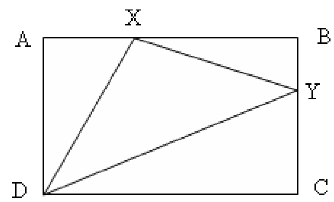
Kalia : I lies on Saturday.
Kaluram : I will lie tomorrow.
Kalia : I lie on Saturday.

On which day of the week did this conversation took place.

7. .

(a) A swimmer is swimming up a river against the current. Next to bridge A he loses an empth water bottle. Twenty minutes later he notices the loss and turns around. He finds the water bottle by bridge B. Knowing that the distance between the two bridges is 2 km. How fast the river flowing?

(b) ABCD is a rectangle and line DX and DY and XY are drawn where X is on AB and Y is on BC. The area of the triangle AXD is 5, the area of triangle BXY is 4 and area of CYD is 3. Determine the area of triangle DXY.



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